

Question Paper

Exam Date & Time: 22-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. EXAMINATION - NOVEMBER 2025

Database Technologies [CSS_5115]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Describe the key roles and responsibilities of a Database Administrator (DBA) in managing a database system. (3)
- A) Course Outcome: "CO1" / Cognitive Level: "Remember"
- B) Design and draw an Entity-Relationship (ER) diagram for a college database with entities Student, Course, and Enrolment, clearly indicating keys and relationships. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Identify and explain four advantages of using a database system instead of traditional flat file storage for managing employee data. (4)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Design a schema diagram for an e-commerce database showing relationships among Customers, Orders, and Products. Identify primary and foreign keys. (4)
- A) Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Write SQL queries to create a table, insert sample data, and perform a join to display customer names and their ordered products. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Scenario: A company wants to restrict duplicate employee IDs and ensure salary is non-negative. Write SQL constraints and explain them. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Explain the three-tier architecture of a data warehouse with a neat labelled diagram. Also, write a basic structure of a PL/SQL block. (3)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Write and explain a PL/SQL function that increases the price of all books in the BOOKS(book_id, title, price) table by 10% where the current price is below ₹500. (3)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Design and draw an ER diagram for a banking system with Customers, Accounts, and Transactions, clearly showing entity relationships and key attributes. Explain how this can be migrated to data warehouse architecture. (4)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) The table STUDENT(id, name, marks) is given. (4)
Write SQL/XML query output to produce a list of < Student> elements containing nested < Name> and < Marks> tags.

A)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- B) A retail company wants to analyze sales by product, region, and quarter. (3)
Design a Star Schema showing fact and dimension tables.

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) What is a Snowflake Schema? How is it different from a Star Schema? (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Explain, with suitable examples, the aggregate data model used in NoSQL databases. (3)
Describe the four main types of NoSQL databases and differentiate their data structures and use cases.

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Demonstrate, with an example, how JSON or XML can be used to transfer data between two applications that use different databases. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) In Brewer's CAP theorem, explain why all three properties of CAP - Consistency, Availability, and Partition Tolerance - cannot be guaranteed simultaneously. (4)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	11
2	2	CO 2	9
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5115:Database Technologies

#	Level	No of Questions	Marks (%)
1	Remember	1	3(6%)

2	Understand	2	8(16%)
3	Application	7	22(44%)
4	Analysis	5	17(34%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)